Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: The Order of Operations in Algebra

When you simplify an algebraic expression, you must work through it in a fixed order, not just left to right. The rule to remember is: **multiply and divide before you add and subtract**, and always deal with anything inside **brackets** first.

Here is how the order goes:

1. Work out anything inside **brackets** first.
2. Then do **multiplication** and **division** from left to right.
3. **Last** do **addition** and **subtraction** from left to right.

You already know how to add and subtract like terms, and how to multiply and divide them. This section is about putting those skills together in the right order so you get one correct answer.

Example 1 — numbers first, to see the order

Simplify (a) $18 - 6 \times 2$, (b) $6 - 18 \div 3$, (c) $6 \times 2 - 2 - 40 \div 4$.

(a) $18 - 6 \times 2$

Multiply before subtracting:
$$= 18 - 12$$
$$= 6$$

(b) $6 - 18 \div 3$

Divide before subtracting:
$$= 6 - 6$$
$$= 0$$

(c) $6 \times 2 - 2 - 40 \div 4$

Do the multiplication and division first:
$$= (6 \times 2) - 2 - (40 \div 4)$$
$$= 12 - 2 - 10$$
$$= 10 - 10$$
$$= 0$$

Example 2 — the same order with letters

Simplify (a) $3x \times 2 + 5x$, (b) $17y - 5 \times 2y$.

(a) $3x \times 2 + 5x$

Multiply first:
$$= 6x + 5x$$

Then add the like terms:
$$= 11x$$

(b) $17y - 5 \times 2y$

Multiply first:
$$= 17y - 10y$$

Then subtract the like terms:
$$= 7y$$

Example 3 — brackets come first

Simplify (a) $7x \div x + 5$, (b) $4x - (6 \div 3)$, (c) $(24x \div 6) + x + (1 \times 5x)$.

(a) $7x \div x + 5$

Divide first. Since $7x \div x = 7$:
$= 7 + 5$
$= 12$

(b) $4x - (6 \div 3)$

Work out the bracket first: $6 \div 3 = 2$.
$= 4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

Do each bracket first. $24x \div 6 = 4x$, and $1 \times 5x = 5x$.
$= 4x + x + 5x$

Now add the like terms from left to right:
$= 5x + 5x$
$= 10x$

Practice

1. Simplify $4n + 3n \times 10$.
2. Simplify $32 + 8x \div 2$.
3. Simplify $3 \times 5x + 4x \div 2$.
4. Simplify $36y \div 6 + y + 2 \times 4y$.

Section 2: Removing Brackets and Simplifying

When an algebraic expression has brackets in it, you can often make it simpler by removing them. Removing a bracket means writing the terms inside it without the bracket, but the sign right before the bracket tells you what to do.

Keep these hints in mind:

- If a bracket has terms inside it, see whether you can simplify those terms first.
- A + sign in front of a bracket does not change the signs of the terms inside. When you remove the bracket, every term keeps its own sign.
- A − sign in front of a bracket changes every sign inside it: each + becomes − and each − becomes +.

After removing the brackets, collect the like terms together and add or subtract them as you already know how.

Let us see the effect of a + sign and a − sign in front of a bracket.

Simplify:
(a) $a + (b + c)$
(b) $a + (b - c)$
(c) $2a - (3b - c)$

Solution

(a) The + before the bracket leaves the signs alone, so
$a + (b + c) = a + b + c.$

(b) Again a + sits before the bracket, so the signs stay the same:
$a + (b - c) = a + b - c.$

(c) Here a − sign is before the bracket, so every sign inside changes: $3b$ is positive inside, and it becomes $-3b$; c is negative inside, and it becomes $+c$. So
$2a - (3b - c) = 2a - 3b + c.$

Now remove the brackets and then collect like terms.

Simplify:
(a) $a + (a - b)$
(b) $6x + (3 - x)$
(c) $(3a + 2) + (5a + a)$

Solution

(a) Remove the bracket (the + keeps the signs), then collect the a terms:
$$a + (a - b) = a + a - b = 2a - b.$$

(b) Remove the bracket, then gather the x terms:
$$6x + (3 - x) = 6x + 3 - x = 6x - x + 3 = 5x + 3.$$

(c) Remove the bracket, then collect the a terms:
$$(3a + 2) + (5a + a) = 3a + 2 + 5a + a = 3a + 5a + a + 2 = 9a + 2.$$

A $-$ sign before a bracket changes every sign inside it. Watch for this in each step.

Simplify:
(a) $5x - (3y + z)$
(b) $10d - (9c - 4d)$
(c) $(6x - y) - (4x - 5y)$

Solution

(a) The $-$ before the bracket changes $+3y$ into $-3y$ and $+z$ into $-z$:
$$5x - (3y + z) = 5x - 3y - z.$$

(b) The $-$ changes $+9c$ into $-9c$ and $-4d$ into $+4d$:
$$10d - (9c - 4d) = 10d - 9c + 4d = 10d + 4d - 9c = 14d - 9c.$$

(c) Remove the first bracket as it is (the + in front of it changes nothing), and change the signs inside the second bracket because a $-$ stands before it:
$$(6x - y) - (4x - 5y) = 6x - y - 4x + 5y = 6x - 4x + 5y - y = 2x + 4y = 2(x + 2y).$$

Practice

1. Remove the brackets and simplify: $(12a - 7) + (4 - 5a)$
2. Remove the brackets and simplify: $(8x - 2y) - (9x - 4y)$
3. Remove the brackets and simplify: $12 - (7 - 2f) - (3 - f)$
4. Remove the brackets and simplify: $(8a - 5) + (14 - 3a)$

Section 3: Word Problems with Algebraic Expressions

A word problem often describes a quantity in words and asks you to write it with a letter, then combine the parts. The key is to let each letter stand for one thing, read the problem slowly to decide whether to add or subtract, and then simplify the expression you build.

Think of it like this: the problem names some objects (a pen, a rubber, the boys and girls in a class) and tells you how their costs or numbers are related. You turn each sentence into algebra, one piece at a time, and then put the pieces together. As you go, keep the naira sign (₦) with every cost, and when you are asked for a total, add all the parts and simplify by collecting like terms.

Example 1

A pen costs ₦P and a pencil costs ₦15 less than a pen.

(a) Find the cost of a pencil.
(b) Find the cost of a pen and a pencil.

Solution

A pencil costs 15 naira less than the pen, so take 15 away from P:

(a) Cost of a pencil = ₦(P − 15).

(b) A pen and a pencil together means we add the two costs:

Cost of pen + cost of pencil = ₦P + ₦(P − 15)
= ₦P + ₦P − ₦15
= ₦2P − ₦15
= ₦(2P − 15).

So a pen and a pencil together cost ₦(2P − 15).

Example 2

A pencil costs ₦x and a rubber costs ₦40 more than a pencil.

(a) Find the cost of a rubber.
(b) Find the cost of a pencil and a rubber.

Solution

A rubber costs 40 naira more than the pencil, so add 40 to x:

(a) Cost of a rubber = ₦(x + 40).

(b) Cost of a pencil and a rubber:

₦x + ₦(x + 40)
= ₦(x + x + 40)
= ₦(2x + 40).

So a pencil and a rubber together cost ₦(2x + 40).

Example 3

In a class there are x boys and y girls. This semester 5 boys join and 3 girls leave the class. How many students are now in the class?

Solution

Start with the total: total number of students = number of boys + number of girls.

Boys: 5 join, so the number of boys becomes x + 5.
Girls: 3 leave, so the number of girls becomes y − 3.

Total number = (x + 5) + (y − 3)
= x + 5 + y − 3
= x + y + 5 − 3
= x + y + 2.

So there are now x + y + 2 students in the class.

Practice

1. A pen costs ₦h and a crayon costs ₦20 less than a pen.
   (a) Find the cost of a crayon.
   (b) Find the cost of a crayon and a pen.
2. Envelopes cost ₦d per pack, and a notepad costs ₦20 more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.
3. A pen and a pencil together cost ₦x. If a pen costs ₦y, find the cost of 3 pencils.

Check your work

Attempt every question before you read this page.

The Order of Operations in Algebra

1. $34n$

   Multiply first: $3n \times 10 = 30n$. Then add the like terms: $4n + 30n = 34n$.
2. $32 + 4x$

   Divide first: $8x \div 2 = 4x$. 32 and $4x$ are not like terms, so they cannot be combined. The answer is $32 + 4x$.
3. $17x$

   Multiply and divide first: $3 \times 5x = 15x$ and $4x \div 2 = 2x$. Then add the like terms: $15x + 2x = 17x$.
4. $15y$

   Divide and multiply first: $36y \div 6 = 6y$ and $2 \times 4y = 8y$. Now add the like terms from left to right: $6y + y = 7y$, then $7y + 8y = 15y$.

Removing Brackets and Simplifying

1. $7a - 3$

   The + before the bracket keeps the signs: $12a - 7 + 4 - 5a$. Collect like terms: $12a - 5a - 7 + 4 = 7a - 3$.
2. $-x + 2y$

   The first bracket keeps its signs; the − before the second bracket changes $+9x$ to $-9x$ and $-4y$ to $+4y$. So $8x - 2y - 9x + 4y = 8x - 9x + 4y - 2y = -x + 2y$.
3. $3f + 2$

   Each − before a bracket changes every sign inside. So $12 - 7 + 2f - 3 + f$. Collect the numbers and the f terms: $12 - 7 - 3 + 2f + f = 2 + 3f = 3f + 2$.
4. $5a + 9$

   The + before the bracket keeps the signs: $8a - 5 + 14 - 3a$. Collect like terms: $8a - 3a + 9 = 5a + 9$.

Word Problems with Algebraic Expressions

1. (a) ₦(h − 20); (b) ₦(2h − 20)

   A crayon costs 20 naira less than the pen, so cost of a crayon = ₦(h − 20). A crayon and a pen together = ₦(h − 20) + ₦h = ₦h − ₦20 + ₦h = ₦2h − ₦20 = ₦(2h − 20).
2. ₦(2d + 20)

   A pack of envelopes costs ₦d. A notepad costs 20 naira more, so it costs ₦(d + 20). Total = ₦d + ₦(d + 20) = ₦(d + d + 20) = ₦(2d + 20).
3. ₦3(x − y)

   Out of the total ₦x, the pen costs ₦y, so one pencil costs ₦x − ₦y = ₦(x − y). Three pencils cost ₦3 × (x − y) = ₦3(x − y).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-d510ff8d	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____